Class Notes/B.Tech/Semester-I/2024-2025

Linear Algebra & Differential Equations (LADE)

UNIT-II : Fundamentals of Vector Spaces and Subspaces

Vector spaces have a wide range of applications in real life. Here are few examples:

1. **Computer graphics**: Vector spaces are used to represent images, animations, and 3D models in computer graphics. For example, a 3D model of a car can be represented as a collection of vectors that describe its shape and position.
2. **Machine learning (**Machine learning is a core component of AI, and it involves training machines to learn from data**):** Vector spaces are used to represent data in machine learning algorithms. For example, an image can be represented as a vector of pixel values, and a text document can be represented as a vector of word frequencies.
3. **Engineering**: Vector spaces are used in engineering to represent physical systems such as electrical circuits and mechanical systems. For example, the behaviour of an electrical circuit can be modelled using vectors that represent the voltages and currents at different points in the circuit.

Important properties of linear systems can be described with the concept and notation of vectors.

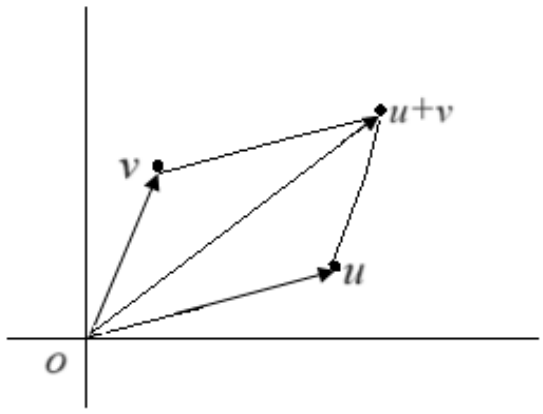
Vectors: A matrix with only one column is called a column vector, or simply a vector. A vector, say u, with n entries $x_1, x_2, x_3, ..., x_n \in R$ is denoted as

$$u = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} \quad or \quad u = (x_1, x_2, x_3, ..., x_n)$$

Vectors in R^2: The set of all vectors with two real entries is denoted by R^2, that is

$$R^2 = \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : x_1, x_2 \in R \right\}$$

Geometric descriptions of R^2: A vector with two real entries can be identified as a geometric point in the co-ordinate plane with arrow from the origin. If $u, v \in R^2$ are represented as points in the plane, then $u+v$ represents to the fourth vector of the parallelogram whose other vertices are u, o, and v as shown below:



Equality of two vectors: Two vectors are equal if and only if their corresponding entries are equal.

Thus, $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ implies that $x = 2,\ y = 3$.

Addition of two vectors in R^2: Let $u = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$ and $v = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$ are two vectors in R^2, then $u+v$ is defined as $u+v = \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \end{bmatrix}$.

Multiplication of a vector with a scalar: Let $u = \begin{bmatrix} x \\ y \end{bmatrix}$ be a vector in R^2 and $c \in R$ be a scalar, then cu is defined as $cu = \begin{bmatrix} cx \\ cy \end{bmatrix}$.

Zero vector: If all entries of a vector are zero, then that vector is called zero vector.

Example: $u_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $u_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Question: Find $2u + 3v - w$, if $u = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $v = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$, and $w = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$.

Solution: $2u+3v-w=2\begin{bmatrix}1\\4\end{bmatrix}+3\begin{bmatrix}-3\\2\end{bmatrix}-\begin{bmatrix}0\\5\end{bmatrix}$

$$=\begin{bmatrix}2\\8\end{bmatrix}+\begin{bmatrix}-9\\6\end{bmatrix}-\begin{bmatrix}0\\5\end{bmatrix}$$

$$=\begin{bmatrix}2-9+0\\8+6-5\end{bmatrix}$$

$$=\begin{bmatrix}-7\\9\end{bmatrix}$$

Vectors in R^3: The set of all vectors with three real entries is denoted by R^3, that is

$$R^3=\left\{\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}: x_1,x_2,x_3\in R\right\}$$

Examples: $u_1=\begin{bmatrix}2\\0\\1\end{bmatrix}$, $u_2=\begin{bmatrix}3\\4\\7\end{bmatrix}$, $u_3=\begin{bmatrix}1\\-3\\0\end{bmatrix}$

Geometrically, vector in R^3 represents a point in three-dimensional co-ordinate space with arrow from the origin.

Vectors in R^n: The set of all vectors with n real entries is denoted by R^n, that is

$$R^n=\left\{\begin{bmatrix}x_1\\x_2\\x_3\\\vdots\\x_n\end{bmatrix}: x_1,x_2,x_3,\cdots,x_n\in R\right\}$$

Vector Space and Subspace

Vector Space:

A vector space V is a non-empty set of vectors, on which two operations called addition and scalar multiplication (real number) are defined, subject to the ten axioms (or rules) listed below.

The axioms must hold for all vectors u, v and w in V and for all real scalars c and d.

1. The sum $u+v$ is in the vector space V (Closure w. r to addition)
2. $u+v=v+u$ (Commutative)
3. $(u+v)+w=u+(v+w)$ (Associative w. r to addition)
4. There exists a zero vector 0 in V, such that $0+u=u$ (Additive identity)
5. For each u, we can find '$-u$' in V, such that $u+(-u)=0$ (Additive inverse)
6. The scalar multiple cu is in the vector space V (Closure w. r to multiplication)
7. $c(u+v)=cu+cv$ (Distributive of multiplication over vector addition)
8. $(c+d)u=cu+du$ (Distributive of multiplication over scalar addition)
9. $c(du)=(cd)u$ (Associative w. r to multiplication)
10. $1.u=u$ (Multiplicative identity)

Examples of Vector Space:

(1) The spaces R^n, where $n \geq 1$, are the premier examples of vector spaces. The geometric intuition developed for R^n will help you understand and visualize many concepts throughout the chapter.

(2) For $n \geq 0$, the set P_n of polynomials of degree at most n consists of all polynomials of the form

$$p(t)=a_0+a_1t+a_2t^2+\ldots+a_nt^n \qquad (1)$$

where, the coefficients $a_0, a_1, a_2, \ldots, a_n$ and the variable t are real numbers. The degree of p is the highest power of t in Equation (1) whose coefficient is not zero. If $p(t)=a_0 \neq 0$, the degree of p is called the zero polynomial. The zero polynomial is included in P_n even though its degree for technical reason is not defined.

If P is given by Equation (1) and if $q(t)=b_0+b_1t+\ldots+b_nt^n$, then the sum $p+q$ is defined by

$$(p+q)(t)=p(t)+q(t)$$

$$(p+q)(t)=(a_0+b_0)+(a_1+b_1)t+\ldots+(a_n+b_n)t^n$$

The scalar multiple cp is the polynomial defined by

$$(cp)(t)=cp(t)=ca_0+(ca_1)t+\ldots+(ca_n)t^n$$

These definitions satisfy axioms (1) and (6) because $p+q$ and cp are polynomials of degree less than or equal to n.

Axioms (2), (3) and (7)-(10) follow from properties of the real numbers. Clearly, the zero polynomial acts as the zero vector in Axiom (4).
Finally, $(-1)p$ acts as the negative of p, so axiom (5) is satisfied.

Thus P_n is a vector space.

(3) Let V be the set of all real-valued function defined on a set D (typically, D is the set of real numbers or some interval on the real line).

- Functions are added in the usual way: $f+g$ is the function whose value at t in the domain D is $f(t)+g(t)$.
- Similarly, for a scalar c and an f in V, the scalar multiple cf is the function whose value at t is $cf(t)$.
- For instance, if $D=R$, $f(t)=1+\sin 2t$ and $g(t)=2+0.5t$, then
$$(f+g)t=3+\sin 2t+0.5t \text{ and } (2g)(t)=4+t$$
- Two functions in V are equal if and only if their values are equal for every t in D.
- Hence the zero vector in V is the function that is identically zero, $f(t)=0$ for all t, and the negative of f is $(-1)f$. Axioms 1 and 6 are obviously true, and the other axioms follow from properties of the real numbers, so V is a vector space.

Questions:

1. Let V be the first quadrant in the xy-plane, i.e., let $V=\left\{\begin{bmatrix} x \\ y \end{bmatrix};\ x\geq 0, y\geq 0\right\}$. Then examine V to be a vector space.

   Solution: (a) If u and v are in V, then their entries are non-negative.
   Since a sum of non-negative numbers is non-negative, the vector $u+v$ has non-negative entries.

   Thus, $u+v$ is in V.

   (b) If $u=\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and $c=-1$, then u is in V but cu is not in V.

   Hence, V is not a vector space.

2. Let W be the union of the first and third quadrant in the $xy-$plane. That is,

   Let $$W=\left\{\begin{bmatrix} x \\ y \end{bmatrix};\quad xy\geq 0\right\}$$

   (a) If u is in W and c is any scalar, is cu in W? Why?
   (b) Find specific vectors u and v in W such that $u+v$ is not in W. This is enough to show that W is not a vector space.

Solution: (a) If $u = \begin{bmatrix} x \\ y \end{bmatrix}$ is in W, then the vector $cu = c\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} cx \\ cy \end{bmatrix}$ is in W, because $(cx)(cy) = c^2(xy) \geq 0$, since $xy \geq 0$.

(c) Example: If $u = \begin{bmatrix} -1 \\ -7 \end{bmatrix}$ and $v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, then u and v are in W but $u+v$ is not in W.

Subspace:

A subspace of a vector space V is a subset H of V that has three properties:

(a) The zero vector of V is in H.
(b) H is closed under vector addition. That is, for each u and v in H, the sum $u+v$ is in H.
(c) H is closed under multiplication by scalars. That is, for each u in H and each scalar c, the vector cu is in H.

Observations:

- Every subspace satisfies all the rules of a vector space. So, every subspace is a vector space.
- Zero-subspace:
  The set $H = \{0\}$ is a subspace of any vector space. This is called the trivial subspace.

Determine Subspace:

Question: Is the following set a subspace of R^3?

$$H = \left\{ \begin{bmatrix} c_1 \\ 0 \\ c_2 \end{bmatrix} : c_1, c_2 \in R \right\}$$

Solution: We have $H = \left\{ \begin{bmatrix} c_1 \\ 0 \\ c_2 \end{bmatrix} : c_1, c_2 \in R \right\}$.

(i) H contains the zero vector as we can take $c_1 = c_2 = 0$.

(ii) For all $a_1, a_2, b_1, b_2 \in R$, we have $\begin{bmatrix} a_1 \\ 0 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ 0 \\ b_2 \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ 0 \\ a_2 + b_1 \end{bmatrix}$. This is in H.

So, H is closed under addition.

(iii) For any scalar c, we have $c\begin{bmatrix} a_1 \\ 0 \\ a_2 \end{bmatrix} = \begin{bmatrix} ca_1 \\ 0 \\ ca_2 \end{bmatrix}$. This is also in H. So, H is closed under scalar multiplication.

Since, all the three properties are satisfied. Hence, H is a subspace.

Question: Is the following set a subspace of R^3:

$$H = \left\{ \begin{bmatrix} 7c_1 \\ 4c_1 \\ -2c_1 \end{bmatrix} : c_1 \in R \right\}$$

Solution: (i) H contains the zero vector for $c_1 = 0$.

(ii) For all $a_1, b_1 \in R$, we have $\begin{bmatrix} 7a_1 \\ 4a_1 \\ -2a_1 \end{bmatrix} + \begin{bmatrix} 7b_1 \\ 4b_1 \\ -2b_1 \end{bmatrix} = \begin{bmatrix} 7a_1 + 7b_1 \\ 4a_1 + 4b_1 \\ -2a_1 - 2b_1 \end{bmatrix} = \begin{bmatrix} 7(a_1 + b_1) \\ 4(a_1 + b_1) \\ -2(a_1 + b_1) \end{bmatrix}$

Which is in H. So, H is closed under addition.

(iv) For any scalar c, we have $c\begin{bmatrix} 7a_1 \\ 4a_1 \\ -2a_1 \end{bmatrix} = \begin{bmatrix} 7(ca_1) \\ 4(ca_1) \\ -2(ca_1) \end{bmatrix}$.

Which is in H. So, H is closed under scalar multiplication.

Hence, H is a subspace of R^3.

Question: Determine if the following set a subspace of R^3:

$$H = \left\{ \begin{bmatrix} c_1 \\ 4 \\ -c_1 \end{bmatrix} : c_1 \in R \right\}$$

Solution: H does not contain the zero vector, even if we consider $c_1 = 0$.

Therefore, first property is not satisfied.

Hence, H is not a subspace of R^3.

Question: Is the following set a subspace of R^3:

$$H = \left\{ \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} : c_1, c_2 \in R \right\}$$

Solutions: A vector $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \in R^2$, so $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \notin R^3$. Hence, H is not a subspace of R^3.

Observation:

- The vector space R^2 is not a subspace of R^3 because R^2 is not even a subset of R^3.

- The set $H = \left\{ \begin{bmatrix} s \\ t \\ 0 \end{bmatrix} : s, t \in R \right\}$ is a subset of R^3 that looks and acts like R^2, although it is logically distinct from R^2.

Question: Let H be the set of points inside and on the unit circle in the xy-plane, that is $H = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x^2 + y^2 \le 1 \right\}$. Find a specific example to show that H is not a subspace of R^2.

Solution: If $u = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$ and $c = 4$, then u is in H but cu is not in H, because $4u = 4\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$. Since, $2^2 + 2^2 = 8$ is not less than equal to 1. H is not closed under scalar multiplication. Thus, H is not a subspace of R^2.

Question: Given v_1 and v_2 in a vector space V, let $H = \text{Span}\{v_1, v_2\}$. Show that H is a subspace of V.

Solution: (i) We can make $0.v_1 + 0.v_2 = 0$, therefore the zero vector is in H.

(ii) To show that H is closed under vector addition:

Let $u, v \in H$ and write $u = c_1v_1 + c_2v_2$, $v = c_3v_1 + c_4v_2$.

Now, $u + v = (c_1v_1 + c_2v_2) + (c_3v_1 + c_4v_2) = (c_1 + c_2)v_1 + (c_3 + c_4)v_2$

This implies that, $u + v$ is in H.

(iii) Let c is any scalar, then $cu = c(c_1v_1 + c_2v_2) = (cc_1)v_1 + (cc_2)v_2$. Therefore, cu is in H and is closed under scalar multiplication.

Thus, H is a subspace of V.

Theorem: If $v_1, v_2, ..., v_m$ are in a vector space V, then $\text{Span}\{v_1, v_2, ..., v_m\}$ is a subspace of V.

Question: Let H be the set of all vectors of the form $(a - 3b,\ b - a,\ a,\ b)$, where a and b are arbitrary scalars. That is, let $H = \{(a - 3b,\ b - a,\ a,\ b) : a, b \in R\}$. Show that H is a subspace of R^4.

Solution: The given vector in H can be written as follows:

$$\begin{bmatrix} a - 3b \\ b - a \\ a \\ b \end{bmatrix} = a\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

Or $\begin{bmatrix} a-3b \\ b-a \\ a \\ b \end{bmatrix} = av_1 + bv_2$, where $v_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \end{bmatrix}$.

This shows that $H = \text{Span}\{v_1, v_2\}$. Thus, H is a subspace of R^4.

Practice Problems

Q1: Let $u = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$, $w = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$. Determine whether v is a linear combination of u and w.

Answer: $\begin{bmatrix} -1 \\ 4 \end{bmatrix} = -\frac{1}{2}\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \frac{11}{4}\begin{bmatrix} 0 \\ 2 \end{bmatrix}$.

Q2: Let $u = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} 4 \\ 2 \\ 14 \end{bmatrix}$, $w = \begin{bmatrix} 3 \\ 6 \\ 10 \end{bmatrix}$, and $z = \begin{bmatrix} -1 \\ 8 \\ -5 \end{bmatrix}$. Determine whether z is a linear combination of u, v and w.

Answer: $\begin{bmatrix} -1 \\ 8 \\ -5 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} - 2\begin{bmatrix} 4 \\ 2 \\ 14 \end{bmatrix} + 2\begin{bmatrix} 3 \\ 6 \\ 10 \end{bmatrix}$.

Q3: Let $u = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $w = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$, and $z = \begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix}$. Determine whether v is a linear combination of u, w and z.

Q4: Let $u = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$ and $w = \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix}$. Then, Span $\{u, v\}$ is a plane through the origin in R^3. Is w in that plane?

Q5: Determine whether the vector $v = (3,\ 9,\ -4,\ -2)$ belongs to the space spanned by the vectors $u_1 = (1,\ -2,\ 0,\ 3)$, $u_2 = (2,\ 3,\ 0,\ -1)$ and $u_3 = (2,\ -1,\ 2,\ 1)$.

Answer: $v = u_1 + 3u_2 - 2u_3$

Q6: Determine whether the following set a subspace of R^3?

$$H = \left\{ \begin{bmatrix} c_1 \\ c_2 \\ 0 \end{bmatrix} : c_1, c_2 \in R \right\}$$

Q7: Let $v_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$, $v_3 = \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}$, $w = \begin{bmatrix} 8 \\ 4 \\ 7 \end{bmatrix}$ and $z = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$.

(i) Is w in the subspace spanned by $\{v_1, v_2, v_3\}$? Why?

(ii) Is z in the subspace spanned by $\{v_1, v_2, v_3\}$? Why?

Q8: Let v be the set of all vectors of the form $(s+3t, s-t, 2s-t, 4t)$. Show that v is a subspace of R^4.

Null Space and Column Space

Null Space: The null space of an $m \times n$ matrix A, written as $\text{Nul}\, A$, is the set of all solutions to the homogeneous equation $Ax = O$, where O is zero vector.

i.e., $\text{Nul}\, A = \{x : x \text{ is in } R^n \text{ and } Ax = O\}$

Note: (i) The null space of a matrix is a vector space.

(ii) The null space of an $m \times n$ matrix A is a subspace of R^n.

Question: Let $A = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix}$ and $u = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$. Determine if u belongs to the null space of A.

Solution: The vector u will be in null space of A, if $Au = O$.

Here, $$Au = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 5-9+4 \\ -25+27-2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Thus, u is in $\text{Nul}\, A$.

Question: Let $A = \begin{bmatrix} -3 & 1 & 6 \\ 2 & 4 & -2 \\ 3 & 2 & 0 \end{bmatrix}$ and $u = \begin{bmatrix} 4 \\ -6 \\ 3 \end{bmatrix}$. Determine if u belongs to the null space of A.

Column Space: The column space of an $m \times n$ matrix A, written as $\text{Col}\, A$, is the set of all linear combinations of the columns of A. If $A = [a_1\ a_2\ \ldots\ a_n]$, then

$$\text{Col}\, A = \text{Span}\{a_1, a_2, \ldots, a_n\}$$

Theorem: The column space of an $m \times n$ matrix A is a subspace of R^m.

Question: Find a matrix A such that $W = \text{Col}\, A$, where $W = \left\{ \begin{bmatrix} 6a-b \\ a+b \\ -7a \end{bmatrix} : \ a, b \in R \right\}$.

Solution: We can write

$$W = \left\{ \begin{bmatrix} 6a-b \\ a+b \\ -7a \end{bmatrix} : \ a, b \in R \right\}$$

or $$W = \left\{ \begin{bmatrix} 6a \\ a \\ -7a \end{bmatrix} + \begin{bmatrix} -b \\ b \\ 0 \end{bmatrix} : \ a, b \in R \right\}$$

or $$W = \left\{ a \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} : \ a, b \in R \right\} = \text{Span} \left\{ \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Hence, the desired matrix is $A = \begin{bmatrix} 6 & -1 \\ 1 & 1 \\ -7 & 0 \end{bmatrix}$.

Note: The column space of an $m \times n$ matrix A is all of R^m if and only if the equation $Ax = b$ has a solution for each b in R^m.

Question: Let $A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$. Find a nonzero vector in Nul A.

Solution: Let $x = (x_1, x_2, x_3, x_4)$ be a nonzero vector such that $Ax = O$, where O is zero vector.

The augmented matrix is

$$[A \ \ O] = \begin{bmatrix} 2 & 4 & -2 & 1 & 0 \\ -2 & -5 & 7 & 3 & 0 \\ 3 & 7 & -8 & 6 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 9 & 0 & 0 \\ 0 & 1 & -5 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Thus, if x satisfied $Ax = O$, then $x_1 = -9x_3$, $x_2 = 5x_3$, $x_4 = 0$, and x_3 is free variable.

In particular, for $x_3 = 1$, we have a required nonzero vector as $x = (-9, 5, 1, 0)$.

Question: Let $u = \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix}$, $v = \begin{bmatrix} 3 \\ -1 \\ 3 \end{bmatrix}$, and $A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$.

(i) Determine if u is in Nul A. Could u be in Col A?

(ii) Determine if v is in Col A. Could v be in Nul A?

Solution: (i) The vector u is in Nul A, if Au becomes zero vector.

$$\text{Now, } Au = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Clearly, u is not in Nul A. Also, with four entries, u could not possibly be in Col A, since Col A is a subspace of R^3.

(ii) Reduce $[A \;\; v]$ to an echelon form as follows:

$$[A \;\; v] = \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ -2 & -5 & 7 & 3 & -1 \\ 3 & 7 & -8 & 6 & 3 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ 0 & 1 & -5 & -4 & -2 \\ 0 & 0 & 0 & 17 & 1 \end{bmatrix}$$

Clearly, $Ax = v$ is consistent and therefore v is in Col A. With only three entries, v could not possibly be in Nul A, since Nul A is a subspace of R^4.

Contrast Between Nul A and Col A for an $m \times n$ Matrix A

Nul A	Col A
1. Nul A is a subspace of $\mathbb{R}^n$.	**1.** Col A is a subspace of $\mathbb{R}^m$.
2. Nul A is implicitly defined; that is, you are given only a condition ($A\mathbf{x} = \mathbf{0}$) that vectors in Nul A must satisfy.	**2.** Col A is explicitly defined; that is, you are told how to build vectors in Col A.
3. It takes time to find vectors in Nul A. Row operations on $[\,A \quad \mathbf{0}\,]$ are required.	**3.** It is easy to find vectors in Col A. The columns of A are displayed; others are formed from them.
4. There is no obvious relation between Nul A and the entries in A.	**4.** There is an obvious relation between Col A and the entries in A, since each column of A is in Col A.
5. A typical vector $\mathbf{v}$ in Nul A has the property that $A\mathbf{v} = \mathbf{0}$.	**5.** A typical vector $\mathbf{v}$ in Col A has the property that the equation $A\mathbf{x} = \mathbf{v}$ is consistent.
6. Given a specific vector $\mathbf{v}$, it is easy to tell if $\mathbf{v}$ is in Nul A. Just compute $A\mathbf{v}$.	**6.** Given a specific vector $\mathbf{v}$, it may take time to tell if $\mathbf{v}$ is in Col A. Row operations on $[\,A \quad \mathbf{v}\,]$ are required.
7. Nul $A = \{\mathbf{0}\}$ if and only if the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.	**7.** Col $A = \mathbb{R}^m$ if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b}$ in $\mathbb{R}^m$.

Linear Combination and Linear Independence of vectors

Linear combination: Let $u_1, u_2, u_3, \ldots, u_m \in R^n$ are m vectors each with n entries, then $v \in R^n$ is a linear combination of $u_1, u_2, u_3, \ldots, u_m$, if there exist scalars $c_1, c_2, c_3, \ldots, c_m \in R$ such that

$$v = c_1u_1 + c_2u_2 + c_3u_3 + \ldots + c_nu_n$$

Question 1: Let $u = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $w = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$. Determine whether w is a linear combination of u and v.

Solution: If possible, let $w = c_1u + c_2v$, where c_1 and c_2 are scalars.

Then, we can write $\begin{bmatrix} 5 \\ 3 \end{bmatrix} = c_1\begin{bmatrix} 2 \\ 3 \end{bmatrix} + c_2\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2c_1 \\ 3c_1 \end{bmatrix} + \begin{bmatrix} -c_2 \\ 2c_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2c_1 - c_2 \\ 3c_1 + 2c_2 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} 2c_1 - c_2 &= 5 \\ 3c_1 + 2c_2 &= 3 \end{aligned}$$

Solving the above two equations for c_1 and c_2 using Gauss elimination method, we get $c_1 = \frac{13}{7}$ and $c_2 = \frac{-9}{7}$.

Hence, $\begin{bmatrix} 5 \\ 3 \end{bmatrix} = \frac{13}{7}\begin{bmatrix} 2 \\ 3 \end{bmatrix} - \frac{9}{7}\begin{bmatrix} -1 \\ 2 \end{bmatrix}$.

Question 2: Let $u = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$, $w = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$, and $z = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$. Determine whether z is a linear combination of u, v and w.

Solution: If possible, let $z = c_1u + c_2v + c_3w$, where c_1, c_2 and c_3 are scalars.

Then, we can write $\begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = c_1\begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} + c_2\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} + c_3\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} c_1 \\ 3c_1 \\ 2c_1 \end{bmatrix} + \begin{bmatrix} -c_2 \\ 0 \\ 2c_2 \end{bmatrix} + \begin{bmatrix} 2c_3 \\ 3c_3 \\ 4c_3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} c_1 - c_2 + 2c_3 \\ 3c_1 + 3c_3 \\ 2c_1 + 2c_2 + 4c_3 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} c_1 - c_2 + 2c_3 &= 4 \\ 3c_1 + 3c_3 &= 5 \\ 2c_1 + 2c_2 + 4c_3 &= 6 \end{aligned}$$

Solving the above system of three equations for c_1, c_2 and c_3 using Gauss elimination method, we get $c_1 = -\frac{1}{6}$, $c_2 = -\frac{1}{2}$ and $c_3 = \frac{11}{6}$.

Hence, $\begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = -\frac{1}{6}\begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} + \frac{11}{6}\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$.

Question 3: In R^4, is $v = \begin{bmatrix} 2 \\ -1 \\ 3 \\ 0 \end{bmatrix}$ a linear combination of

$$v_1 = \begin{bmatrix} 1 \\ 3 \\ 2 \\ 0 \end{bmatrix}, \; v_2 = \begin{bmatrix} 5 \\ 1 \\ 2 \\ 4 \end{bmatrix} \text{ and } v_3 = \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}?$$

Solution: If possible, we write linear combination as

$$v = c_1 v_1 + c_2 v_2 + c_3 v_3$$

which is a vector equation, where c_1, c_2, c_3 are scalars.

To find c_1, c_2, c_3, we proceed as follows:

$$c_1 \begin{bmatrix} 1 \\ 3 \\ 2 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 5 \\ 1 \\ 2 \\ 4 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \\ 0 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} c_1 \\ 3c_1 \\ 2c_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 5c_2 \\ c_2 \\ 2c_2 \\ 4c_2 \end{bmatrix} + \begin{bmatrix} -c_3 \\ 0 \\ 2c_3 \\ c_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \\ 0 \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} c_1 + 5c_2 - c_3 \\ 3c_1 + c_2 + 0.c_3 \\ 2c_1 + 2c_2 + 2c_3 \\ 0.c_1 + 4c_2 + c_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \\ 0 \end{bmatrix}$$

$$\Rightarrow \quad \begin{aligned} c_1 + 5c_2 - c_3 &= 2 \\ 3c_1 + c_2 + 0.c_3 &= -1 \\ 2c_1 + 2c_2 + 2c_3 &= 3 \\ 0.c_1 + 4c_2 + c_3 &= 0 \end{aligned}$$

which is a system of four linear equations.

Now,

$$[A:B] = \begin{bmatrix} 1 & 5 & -1 & 2 \\ 3 & 1 & 0 & -1 \\ 2 & 2 & 2 & 3 \\ 0 & 4 & 1 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 5 & -1 & 2 \\ 0 & -14 & 3 & -7 \\ 0 & -8 & 4 & -1 \\ 0 & 4 & 1 & 0 \end{bmatrix}, \begin{bmatrix} R_2 \to R_2 - 3R_1 \\ R_3 \to R_3 - 2R_1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 5 & -1 & 2 \\ 0 & -14 & 3 & -7 \\ 0 & 0 & 32 & 42 \\ 0 & 0 & 26 & -28 \end{bmatrix}, \begin{bmatrix} R_3 \to 14R_3 - 8R_2 \\ R_4 \to 14R_4 + 4R_2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 5 & -1 & 2 \\ 0 & -14 & 3 & -7 \\ 0 & 0 & 16 & 21 \\ 0 & 0 & 0 & -721 \end{bmatrix}$$

Clearly, the system is inconsistent and has no solution. Therefore, v cannot be expressed as a linear combination of v_1, v_2, v_3.

Questions 4: Let $u = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$, $w = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$. Determine whether v is a linear combination of u and w.

Answer: $\begin{bmatrix} -1 \\ 4 \end{bmatrix} = -\frac{1}{2}\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \frac{11}{4}\begin{bmatrix} 0 \\ 2 \end{bmatrix}$.

Question 5: Let $u = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} 4 \\ 2 \\ 14 \end{bmatrix}$, $w = \begin{bmatrix} 3 \\ 6 \\ 10 \end{bmatrix}$, and $z = \begin{bmatrix} -1 \\ 8 \\ -5 \end{bmatrix}$. Determine whether z is a linear combination of u, v and w.

Answer: $\begin{bmatrix} -1 \\ 8 \\ -5 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} - 2\begin{bmatrix} 4 \\ 2 \\ 14 \end{bmatrix} + 2\begin{bmatrix} 3 \\ 6 \\ 10 \end{bmatrix}$.

Question 6: Let $u = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $w = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$, and $z = \begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix}$. Determine whether v is a linear combination of u, w and z.

Vector equation: Let a system of linear equations is

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}$$

Denoting $u_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix}$, $u_2 = \begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \end{bmatrix}$, $u_3 = \begin{bmatrix} a_{13} \\ a_{23} \\ a_{33} \end{bmatrix}$, and $v = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

Then, the given system of linear equations is represented in vector equation form as follows:

$$x_1u_1 + x_2u_2 + x_3u_3 = v$$

To find the solution of the above vector equation, we can directly proceed with the augmented matrix $[u_1 \ u_2 \ u_3 \ v]$ using any suitable method, like Gauss elimination method etc.

Span of vectors: If $v_1, v_2, v_3, ..., v_m \in R^n$, then the set of all linear combinations of $v_1, v_2, v_3, ..., v_m$ is denoted by Span$\{v_1, v_2, v_3, ..., v_m\}$ and is called the subset of R^n spanned (or generated) by $v_1, v_2, v_3, ..., v_m$. That is, Span$\{v_1, v_2, v_3, ..., v_m\}$ is the collection of all vectors that can be written in the form

$$c_1v_1 + c_2v_2 + ... + c_mv_m, \text{ where } c_1, c_2, ..., c_m \text{ are scalars.}$$

Note: (i) If we take $c_1 = c_2 = ... = c_m = 0$, then the linear combination $c_1v_1 + c_2v_2 + ... + c_mv_m$ becomes a zero vector. Hence, zero vector is a member of Span$\{v_1, v_2, v_3, ..., v_m\}$.

(ii) Span of a vector, say v in R^3 i.e., Span$\{v\}$ represents a line through the origin. Again, Span of two vectors, say u and v in R^3 i.e., Span$\{u, v\}$ represents a plane through the origin.

Question: Let $u_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$, $u_2 = \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix}$, and $v = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}$. Then, Span$\{u_1, u_2\}$ is a plane through the origin in R^3. Is v in that plane?

Solution: The vector v lies in that plane, if it is possible to find scalars c_1 and c_2 such that

$$c_1u_1 + c_2u_2 = v.$$

Here, $c_1 \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix} = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}$

$$\Rightarrow \begin{aligned} c_1 + 5c_2 &= -3 \\ -2c_1 - 13c_2 &= 8 \\ 3c_1 - 3c_2 &= 1 \end{aligned}$$

Nor the augmented matrix is

$$[A:B] \sim \begin{bmatrix} 1 & 5 & -3 \\ -2 & -13 & 8 \\ 3 & -3 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & -18 & 10 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$

Clearly the system is not consistent. Hence, the vector equation $c_1u_1 + c_2u_2 = v$ has no solution, and so v is not in Span $\{u, v\}$.

Question: Let $u = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$ and $w = \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix}$. Then, Span $\{u, v\}$ is a plane through the origin in R^3. Is w in that plane?

Question: Determine whether the vector $v = (3,\ 9,\ -4,\ -2)$ belongs to the space spanned by the vectors $u_1 = (1,\ -2,\ 0,\ 3)$, $u_2 = (2,\ 3,\ 0,\ -1)$ and $u_3 = (2,\ -1,\ 2,\ 1)$.

Answer: $v = u_1 + 3u_2 - 2u_3$

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Linearly Independent Vectors: Let $v_1, v_2, v_3, ..., v_m \in V$. The vectors $v_1, v_2, v_3, ..., v_m$ are said to be linearly independent if the vector equation

$$c_1v_1 + c_2v_2 + c_3v_3 + ... + c_mv_m = O$$

has only the trivial solution i.e., $c_1 = c_2 = c_3 = ... = c_m = 0$, where $c_1, c_2, ..., c_m$ are scalars and O is zero vector.

Note: (i) If the above vector equation has a nontrivial solution i.e., at least one scalar among $c_1, c_2, ..., c_m$ is not zero, then the vectors $v_1, v_2, v_3, ..., v_m$ are called **linearly dependent vectors**.

(ii) If vectors are linearly dependent, then one vector can be written as a linear combination of the others remaining vectors.

Question: Examine the linear dependency of the two vectors $u = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

Solution: Consider the vector equation as

$$c_1 u + c_2 v = O \qquad (1)$$

where, c_1 and c_2 are scalars and O is zero vector.

Solving the vector equation for c_1 and c_2 as follows:

$$c_1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -c_1 \\ c_1 \end{bmatrix} + \begin{bmatrix} 2c_2 \\ 3c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -c_1 + 2c_2 \\ c_1 + 3c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} -c_1 + 2c_2 &= 0 \\ c_1 + 3c_2 &= 0 \end{aligned}$$

Solving the above system of two linear equations, we get $c_1 = c_2 = 0$. Clearly, the given vectors u and v are linearly independent.

Note: To find the solution of the above vector equation

$$c_1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

we can directly proceed as follows:

$$\left[\begin{array}{cc|c} -1 & 2 & 0 \\ 1 & 3 & 0 \end{array}\right]$$

$$\sim \left[\begin{array}{cc|c} -1 & 2 & 0 \\ 0 & 5 & 0 \end{array}\right], \quad [R_2 \to R_2 + R_1]$$

The linear system corresponding to the vector equation (1) is consistent and has unique solution. The solution is given by back substitution as follows:

$$\begin{aligned} -c_1 + 2c_2 &= 0 \\ 5c_2 &= 0 \end{aligned}$$

$$\Rightarrow \quad c_1 = c_2 = 0.$$

Question: Examine the linear dependency of the two vectors $u = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ -1/3 \end{bmatrix}$.

Solution: Consider the vector equation as

$$c_1 u + c_2 v = O \qquad (1)$$

where, c_1 and c_2 are scalars and O is zero vector.

Solving the vector equation for c_1 and c_2 as follows:

$$c_1 \begin{bmatrix} -3 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1/3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Or $$\left[\begin{array}{cc|c} -3 & 1 & 0 \\ 1 & -1/3 & 0 \end{array}\right]$$

$$\sim \left[\begin{array}{cc|c} -3 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right], \quad [R_2 \to R_2 + (1/3)R_1]$$

The linear system corresponding to the vector equation (1) is consistent and will have infinitely many solutions. Here c_1 is basic unknown and c_2 is free unknown.

Now, we can write

$$-3c_1 + c_2 = 0 \quad \text{or} \quad c_1 = \frac{1}{3} c_2$$

Here, for different values of c_2, we get different values of c_1. Also, from the vector equation, we can write $u + 3v = 0$. Hence, the given two vectors are linearly dependent.

Question: Determine whether the vectors $u = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$, $v = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$ and $w = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$ in R^3 are linearly dependent or independent.

Solution: Consider the following vector equation

$$c_1 u + c_2 v + c_3 w = O \qquad (1)$$

where, c_1, c_2, c_3 are scalars and O is zero vector.

Solving the vector equation for c_1, c_2, c_3 as follows:

$$\left[\begin{array}{ccc|c} -3 & 0 & 2 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ -3 & 0 & 2 & 0 \\ 0 & -1 & 0 & 0 \end{array}\right], \ [R_1 \leftrightarrow R_2]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & -1 & 0 & 0 \end{array}\right], \ [R_2 \to R_2 + 3R_1]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 2/3 & 0 \end{array}\right], \ [R_3 \to R_3 + (1/3)R_2]$$

Clearly, the system of linear equations corresponding to the Equation (1) in three unknowns c_1, c_2, c_3 is consistent and has unique solution. The corresponding system of reduced equations is

$$c_1 + c_2 = 0$$
$$3c_2 + 2c_3 = 0$$
$$\frac{2}{3}c_3 = 0$$

⇨ $c_1 = c_2 = c_3 = 0$.

Hence, the given vectors u, v and w are linearly independent.

Question:

Theorem: Let $v_1, v_2, v_3, ..., v_m \in R^n$. If at least one of these m vectors is zero vector, then the vectors $v_1, v_2, v_3, ..., v_m$ are linearly dependent vectors.

Example: The vectors $v_1 = \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix}$, $v_2 = \begin{bmatrix} -4 \\ 19 \\ -5 \end{bmatrix}$ and $v_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ are linearly dependent vectors as we can write $0v_1 + 0v_2 + cv_3 = O$ for all non-zero scalars c.

Theorem: Let $v_1, v_2, v_3, ..., v_m \in R^n$. If $n < m$, then the vectors $v_1, v_2, v_3, ..., v_m$ are linearly dependent vectors i.e., if there are more vectors than the value of n in the n-space then the vectors are linearly dependent.

Question: Test the vectors $u=\begin{bmatrix}-3\\1\\0\end{bmatrix}$, $v=\begin{bmatrix}0\\1\\-1\end{bmatrix}$, $w=\begin{bmatrix}2\\0\\0\end{bmatrix}$, and $z=\begin{bmatrix}1\\2\\3\end{bmatrix}$ in R^3 for linear independence.

Solution: Consider the following vector equation

$$c_1u+c_2v+c_3w+c_4z=O \qquad (1)$$

where, c_1, c_2, c_3, c_4 are scalars and O is zero vector.

Solving the vector equation for c_1, c_2, c_3, c_4 as follows:

$$\left[\begin{array}{cccc|c}-3 & 0 & 2 & 1 & 0\\ 1 & 1 & 0 & 2 & 0\\ 0 & -1 & 0 & 3 & 0\end{array}\right]$$

$$\sim\left[\begin{array}{cccc|c}1 & 1 & 0 & 2 & 0\\ -3 & 0 & 2 & 1 & 0\\ 0 & -1 & 0 & 3 & 0\end{array}\right],\ [R_1 \leftrightarrow R_2]$$

$$\sim\left[\begin{array}{cccc|c}1 & 1 & 0 & 2 & 0\\ 0 & 3 & 2 & 7 & 0\\ 0 & -1 & 0 & 3 & 0\end{array}\right],\ [R_2 \leftrightarrow R_2+3R_1]$$

$$\sim\left[\begin{array}{cccc|c}1 & 1 & 0 & 2 & 0\\ 0 & 3 & 2 & 7 & 0\\ 0 & 0 & 2/3 & 16/3 & 0\end{array}\right],\ [R_3 \leftrightarrow R_3+(1/3)R_2]$$

Clearly, the system of linear equations corresponding to the vector equation (1) in four unknowns c_1, c_2, c_3, c_4 is consistent and has infinite many solutions. Here, c_1, c_2, c_3 are basic unknowns and corresponding to the fourth column, c_4 is the free unknown.

The corresponding system of reduced equations is

$$\begin{aligned}&c_1+c_2+2c_4=0\\&3c_2+2c_3+7c_4=0\\&-c_2+3c_4=0\end{aligned}$$

$$\Rightarrow\ c_1=-5c_4,\ c_2=3c_4,\ c_3=-8c_4$$

The unknowns c_1, c_2, c_3 are non-zero for different non-zero values of c_4. Hence, the given vectors are linearly dependent vectors.

Note: In particular, for $c_4=1$, the above given vectors u, v, w and z can be written from vector equation (1) as

$$z = 5u - 3v + 8w$$

Clearly, the vectors are linearly dependent.

Theorem (Invertible matrix theorem): Let $v_1, v_2, v_3, ..., v_n$ are n vectors in R^n each with n entries. Then the matrix $A = [v_1 \;\; v_2 \;\; v_3 \; ... \; v_n]$ is invertible (i.e., A^{-1} exists) if the vectors $v_1, v_2, v_3, ..., v_n$ are linearly independent.

Note: Also note that, if the rank of a square matrix is equal to the order of the matrix, then the inverse of the matrix exits. Whereas, if the rank is less than the order of the matrix, then the inverse of the matrix does not exit.

Practice Problems

Q 1: Examine the linear dependency of the following two vectors:

(i) $u = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $v = \begin{bmatrix} -6 \\ -8 \end{bmatrix}$ or $u = (3, \; 4)$ and $v = (-6, \; -8)$

(ii) $u_1 = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$ and $u_2 = \begin{bmatrix} -3 \\ -5 \end{bmatrix}$

(iii) $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

(iv) $u_1 = (1, \; 3), \; u_2 = (2, \; 5)$

(v) $v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} -2 \\ -2 \\ -2 \end{bmatrix}$

(vi) $u = \begin{bmatrix} -2 \\ 3 \\ 0 \end{bmatrix}, v = \begin{bmatrix} 6 \\ -1 \\ 5 \end{bmatrix}$

Q 2: Examine the linear dependency of the following three vectors:

(i) $u = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}, v = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, w = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

(ii) $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, v = \begin{bmatrix} 3 \\ 5 \end{bmatrix}, w = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

(iii) $e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

(iv) $v_1 = \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ -4 \\ 6 \end{bmatrix}, v_3 = \begin{bmatrix} 2 \\ 0 \\ 6 \end{bmatrix}$

(v) $u = (1,\ 0,\ -2),\ v = (3,\ 2,\ -4),\ w = (-3,\ -5,\ 1)$

Q 3: Examine the linear dependency of the following four vectors:

(i) $v_1 = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 3 \\ -1 \end{bmatrix}, v_3 = \begin{bmatrix} 3 \\ -5 \\ 4 \end{bmatrix}, v_4 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}$

(ii) $u = \begin{bmatrix} 1 \\ -3 \\ 0 \end{bmatrix}, v = \begin{bmatrix} -2 \\ 9 \\ 0 \end{bmatrix}, w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, z = \begin{bmatrix} 0 \\ -3 \\ 5 \end{bmatrix}$

(iii) $e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, e_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

Q 4: Find the value(s) of h for which the following vectors are linearly dependent:

(i) $\begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ -6 \\ 7 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \\ h \end{bmatrix}$

(ii) $\begin{bmatrix} 3 \\ -6 \\ 1 \end{bmatrix}, \begin{bmatrix} -6 \\ 4 \\ -3 \end{bmatrix}, \begin{bmatrix} 9 \\ h \\ 3 \end{bmatrix}$

Basis and Dimension of a Vector Space

Basis: Let H be a subspace of a vector space V. An indexed set of vectors $B = \{v_1, v_2, v_3, ..., v_m\}$ in V is a basis of H if

(i) B is a linearly independent set, and
(ii) the subspace spanned by B coincides with H; that is,

$$H = \text{Span}\{v_1, v_2, v_3, ..., v_m\}$$

Example 1: Let A be an invertible $n \times n$ matrix, say $A = [a_1 \quad a_2 \quad ... \quad a_n]$. Then the columns of A form a basis for R^n because they are linearly independent and they span R^n, by the invertible matrix theorem.

Example 2: Let $e_1, e_2, e_3, ..., e_n$ be the column of the $n \times n$ identity matrix, I_n. That is,

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \ldots, e_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

The set $\{e_1, e_2, e_3, \ldots, e_n\}$ is called the standard basis for R^n.

Question: Let $u = \begin{bmatrix} 3 \\ 0 \\ -6 \end{bmatrix}, v = \begin{bmatrix} -4 \\ 1 \\ 7 \end{bmatrix}, w = \begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix}$. Determine if $\{v_1, v_2, v_3\}$ is a basis for R^3.

The spanning set theorem: Let $S = \{v_1, v_2, v_3, \ldots, v_m\}$ be a set in V, and let $H = \text{Span}\{v_1, v_2, v_3, \ldots, v_m\}$.

(i) If one of the vectors in S, say v_k, is a linear combination of the remaining vectors in S, then the set formed from S by removing v_k still spans H.

(ii) If $H \neq \{0\}$, some subset of S is a basis for H.

Theorem: If a vector space V has a basis $B = \{v_1, v_2, v_3, \ldots, v_m\}$, then any set in V containing more than n vectors must be linearly dependent.

Theorem: If a vector space V has a basis of n vectors, then every basis of V must consist of exactly n vectors.

Dimension of a Vector Space: If V is spanned by a finite set, then V is said to be finite-dimensional, and the dimension of V, written as dim V, is the number of vectors in a basis for V.

Note: (i) The dimensional of the zero vector space $\{0\}$ is defined to be zero.

(ii) If V is not spanned by a finite set, then V is said to be infinite-dimensional.

Example: Let $H = \text{Span}\{v_1, v_2\}$, where $v_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$. Then H represents a plane. A basis for H is $\{v_1, v_2\}$, since v_1 and v_2 are not multiples and hance are linearly independent. Thus dim $H = 2$.

Example: Find the dimension of the subspace

$$H = \left\{ \begin{bmatrix} a-3b+6c \\ 5a+4d \\ b-2c-d \\ 5d \end{bmatrix} : a,b,c,d \in R \right\}$$

Solution: It is easy to see that H is the set of all linear combinations of the vectors

$$v_1 = \begin{bmatrix} 1 \\ 5 \\ 0 \\ 0 \end{bmatrix}, \; v_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \; v_3 = \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \; v_4 = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 5 \end{bmatrix}$$

Clearly, $v_1 \neq 0$, v_2 is not a multiple of v_1, but v_3 is a multiple of v_2. By the spanning set theorem, we may discard v_3 and still have a set that spans H. Finally, v_4 is not a linear combination of v_1 and v_2. So $\{v_1, v_2, v_4\}$ is linearly independent and hence is a basis for H. Thus dim $H = 3$.

Subspace of a Finite-dimensional Space

Theorem: Let H be a subspace of a finite-dimensional vector space V. Any linearly independent set in H can be expanded, if necessary, to a basis for H. Also, H is finite-dimensional and

$$\dim H \leq \dim V$$

Theorem (basis theorem): Let V be a p-dimensional vector space, $p \geq 1$. Any linearly independent set of exactly p elements in V is automatically a basis for V. Any set of exactly p elements that span V is automatically a basis for V.

The Dimensions of Nul *A* and Col *A*: The dimension of Nul A is the number of free variables in the equation $Ax = O$, and the dimension of Col A the number of pivot columns in A.

Example: Find the dimension of the null space and the column space of

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Solution: Row reduce the augmented matrix $[A \mid O]$ to echelon form:

$$A = \left[\begin{array}{ccccc|c} 1 & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

There are three free variables x_2, x_4 and x_5. Hence the dimension of Nul A is 3. Also, dim Col $A = 2$ because A has two pivot columns.